

QUANTUM FOAM

Under extreme conditions, such as in the center of a black hole or an instant after the Big Bang, the tried-and-tested laws of physics (General Relativity and quantum physics) break down.

It is in this realm—the Planck scale—that physicists expect quantum gravity to emerge. The extreme quantities associated with this scale are measured with Planck units such as Planck length ($\sim 10^{-35}$ m). One unit of Planck length in comparison to the size of an atom is similar to the length of a football field relative to the entire visible universe.